From myeloma precursor disease to multiple myeloma: new diagnostic concepts and opportunities for early intervention.

Since monoclonal gammopathy of undetermined significance (MGUS) was first described more than 30 years ago, the definition of the entity has evolved. Today, 3 distinct clinical MGUS subtypes have been defined: non-immunoglobulin M (IgM; IgG or IgA) MGUS, IgM MGUS, and light chain MGUS. Each clinical MGUS subtype is characterized by unique intermediate stages and progression events. Although we now have strong evidence that multiple myeloma is consistently preceded by a precursor state at the molecular level, there is urgent need to better understand mechanisms that regulate transformation from precursor to full-blown multiple myeloma. In the future, if such knowledge was available, it would allow clinicians to define high-risk and low-risk precursor patients for a more tailored clinical management. Also, it would provide insights on the individual patient's disease biology, which, in turn, can be used for targeted and more individualized treatment strategies. On the basis of current clinical guidelines, patients diagnosed with MGUS and smoldering myeloma should not be treated outside of clinical trials. In the near future, it seems reasonable to believe that high-risk precursor patients will likely become candidates for early treatment strategies. In this review, we discuss novel insights from recent studies and propose future directions of relevance for clinical management and research studies.